

Akshay V Gudur

+91 9886543644 | akshayvijaygudur@gmail.com | <https://github.com/sickDoughnut-007> | <https://www.linkedin.com/in/akshay-gudur-570017333/> | leetcode.com/u/akshaygudur

EDUCATION

PES University, RR Campus — Bengaluru, KA

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.76 — Pre-Final Year

EXPERIENCE

Research Intern — Quantum-Resistant Key Transport Pipeline

Centre for Information Security, Forensics and Cyber Resilience (ISFCR), PES University | Cryptography, Steganography

- Designed and implemented a quantum-resistant covert key transport pipeline combining AES-256-GCM, Shamir Secret Sharing (3-of-5 threshold), HMAC-SHA256 authentication, Reed-Solomon FEC, and JPEG DCT domain steganography (F5 algorithm)
- Built and validated a comprehensive evaluation suite spanning imperceptibility (PSNR/SSIM), steganalysis (Aletheia, Stegdetect), and cross-platform performance benchmarking across commodity and embedded (Raspberry Pi) hardware
- Co-authoring findings for submission to IEEE Access and an accompanying textbook chapter on post-quantum cryptography for multimedia security

PROJECTS

PESU I/O — ML Model Optimization / Python, TensorFlow

- Worked on a PESU I/O research project using TensorFlow; team received the Distinction Award
- Implemented and optimized models based on the N-BEATS research paper for time-series forecasting
- Improved model accuracy and response speed by re-implementing architectures from research papers
- Extracted and adapted a novel model architecture and deployed it into the pipeline

Decentralized Smart Traffic Grid / Python, Raspberry Pi, Tailscale VPN, UDP

- Engineered a distributed IoT architecture using a headless Raspberry Pi as an edge node to process local telemetry and execute real-time logic
- Established a secure Zero-Trust telemetry pipeline by routing asynchronous UDP packets through an encrypted Tailscale VPN tunnel
- Developed a multi-threaded Python stress-launcher simulating 100+ concurrent traffic nodes without packet loss
- Built a terminal-based UI using ANSI escape codes for real-time visualization of synchronized traffic states

AI Dermatologist Assistant — Medical AI System / Python, FastAPI, React, MongoDB

- Developed an AI-powered dermatology assistant capable of classifying skin diseases (including skin cancer) from uploaded images using deep learning models
- Built a full-stack system with a React frontend and FastAPI backend for real-time inference and visualization
- Integrated a Gemini API-based chatbot for symptom analysis, condition suggestions, and treatment guidance
- Designed an end-to-end pipeline for preprocessing, model inference, and response generation
- Enabled conversational follow-up queries for enhanced diagnostic interaction

HACKATHONS & COMPETITIONS

Terrathon Hackathon — AI Dermatologist Assistant

- Built an ML-based skin disease and skin cancer detection system from uploaded images
- Integrated a Gemini-API chatbot with prompt engineering for precise symptom analysis and remedies

Cryovault CTF Hackathon

- Solved challenges in cryptography, reverse engineering, and security exploitation
- Worked on encryption techniques, binary analysis, and logical vulnerabilities

Algomania 3.0 — Competitive Programming — 27th Place

- Top 40 out of 118 teams in prelims
- Demonstrated strong algorithmic problem-solving under time constraints

TECHNICAL SKILLS

Languages: C, C++, Python, Java, HTML, JavaScript

Frameworks: TensorFlow, React, Node.js

Libraries: Pandas, NumPy, Matplotlib, SciPy

Developer Tools: Git, MongoDB, Gemini API, Linux/Bash, Tailscale/Networking